Supplementary Information

A Multi-Tissue Transcriptomic Atlas of Porcine Development Identifies Conserved Molecular Programs with Humans

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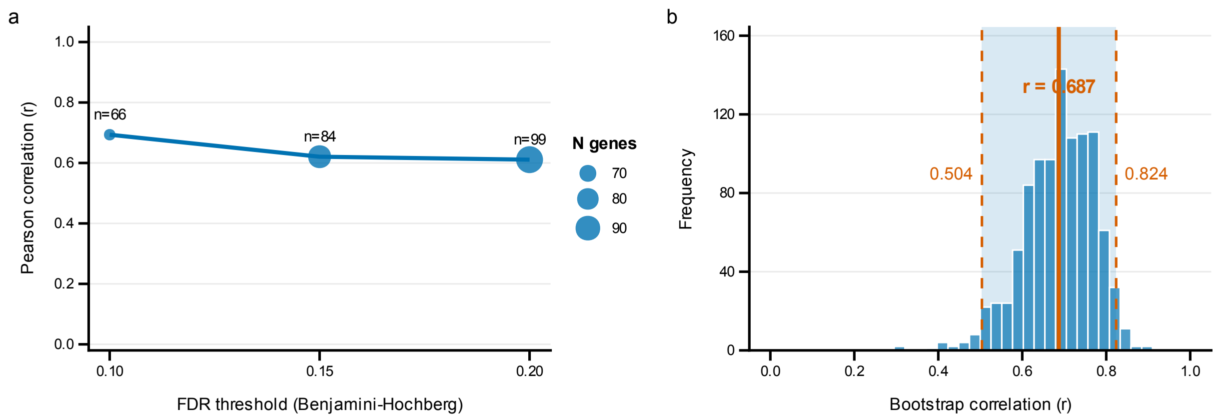
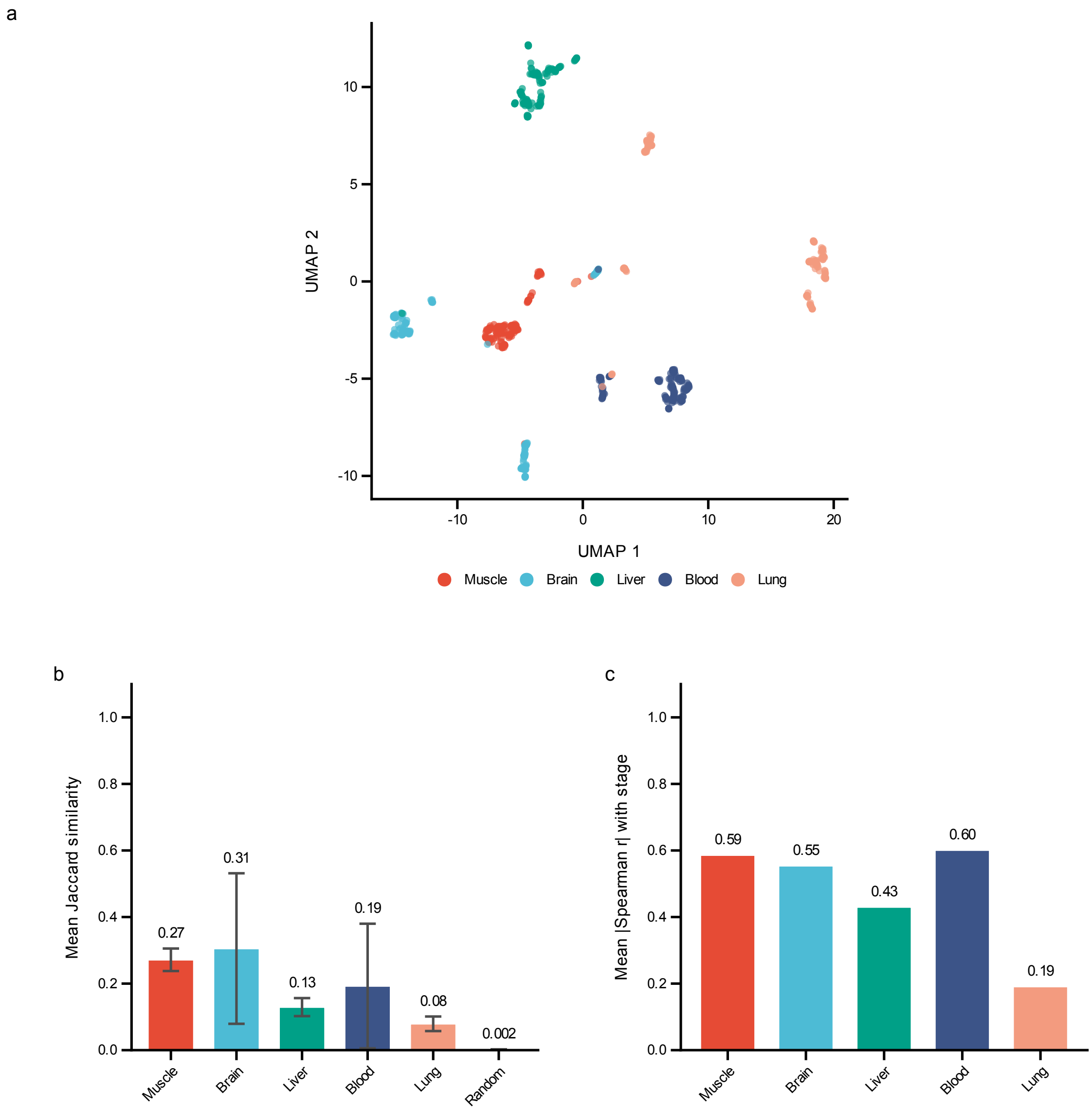
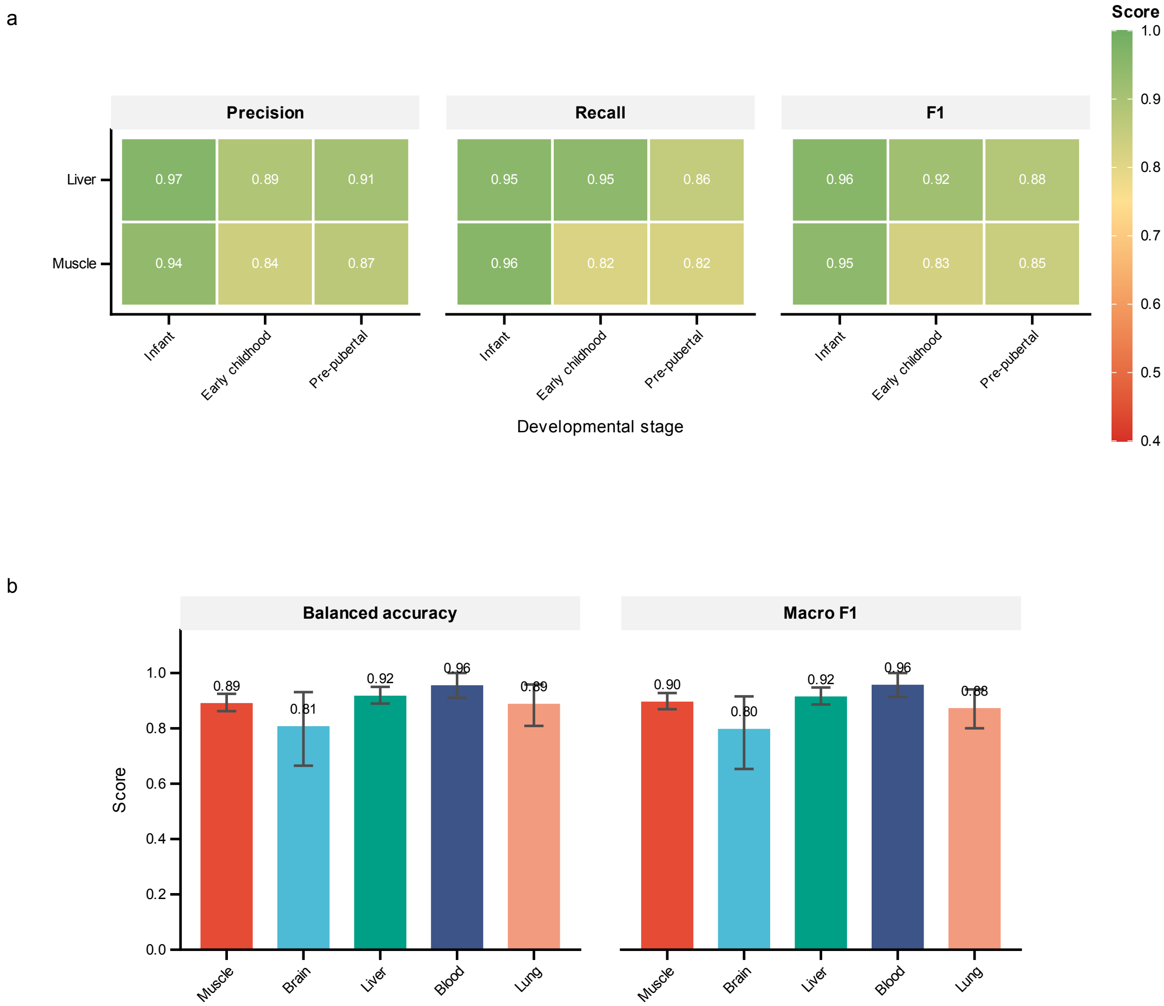
Supplementary Figures



Supplementary Figure 1: Cross-species sensitivity analysis of conserved developmental markers. a, Pearson correlation between pig and human developmental fold changes across FDR thresholds ranging from < 0.05 to < 0.20 (Benjamini–Hochberg corrected), using all genes passing pre-specified criteria (FDR $<$ threshold and $|\log_2 \text{FC}| > 0.5$ in both species). Points are sized by the number of genes at each threshold. **b**, Bootstrap distribution of correlation coefficients ($n = 1{,}000$ iterations) for the pre-specified criteria (FDR < 0.10, $|\log_2 \text{FC}| > 0.5$, $n = 66$ genes). Solid vertical line indicates the observed correlation ($r = 0.693$); dashed lines denote the 95% bootstrap confidence interval boundaries (0.50–0.82).



Supplementary Figure 2: Tissue specificity and feature selection stability analysis. a, Uniform manifold approximation and projection (UMAP) of gene expression profiles across five tissues (muscle, brain, liver, blood, lung), demonstrating distinct tissue-specific clustering patterns ($n = 150$ samples per tissue, balanced sampling; top 500 variable genes). **b**, Within-tissue feature selection stability quantified as mean Jaccard similarity index of the top 50 selected features across ten independent random seeds. Error bars represent standard deviation; higher values indicate more reproducible feature selection. **c**, Mean absolute Spearman correlation between top-ranked features and developmental stage for each tissue.



Supplementary Figure 3: Extended classification performance metrics across tissues and developmental stages. a, Heatmap of per-class precision, recall, and F1 scores for each developmental stage across all five tissues. Colour scale ranges from 0.4 (red, poor performance) to 1.0 (green, excellent performance). Missing values indicate developmental stages not present in that tissue's classification scheme. **b**, Overall performance summary showing balanced accuracy and macro-averaged F1 scores evaluated on the held-out test set (30% of samples) for each tissue. Error bars indicate 95% bootstrap confidence intervals from 1,000 resampling iterations.

Supplementary Tables

Table 1: Train versus test performance comparison. Both train set (70%) and test set (30% held out) performance are reported for all tissues. The gap between train and test metrics indicates the degree of overfitting; regularisation and early stopping were used to minimise this gap.

Tissue	Set	Scheme	n	Balanced Acc.	F1-macro	F1-weighted
Muscle	Train	4-class	761	0.968	0.974	0.984
Muscle	Test	4-class	761	0.894	0.899	0.939
Brain	Train	2-class	43	0.915	0.922	0.930
Brain	Test	2-class	43	0.811	0.801	0.816
Liver	Train	4-class	309	0.985	0.987	0.987
Liver	Test	4-class	309	0.921	0.918	0.919
Blood	Train	2-class	78	1.000	1.000	1.000
Blood	Test	2-class	78	0.959	0.960	0.961
Lung	Train	2-class	129	0.948	0.942	0.961
Lung	Test	2-class	129	0.892	0.876	0.916

Table 2: Per-class performance metrics by tissue. Precision, recall, and F1 scores for each developmental stage within each tissue. n indicates the number of test samples per class.

Tissue	Stage	n	Precision	Recall	F1
Muscle	Infant	194	0.939	0.959	0.949
Muscle	Early childhood	76	0.838	0.816	0.827
Muscle	Pre-pubertal	90	0.871	0.822	0.846
Muscle	Post-pubertal/Adult	401	0.973	0.980	0.976
Brain	Pre-pubertal ($<$150d)	28	0.885	0.821	0.852
Brain	Post-pubertal ($\geq$150d)	15	0.706	0.800	0.750
Liver	Infant	87	0.965	0.954	0.960
Liver	Early childhood	58	0.887	0.948	0.917
Liver	Pre-pubertal	83	0.910	0.855	0.882
Liver	Post-pubertal/Adult	81	0.904	0.926	0.915
Blood	Pre-pubertal ($<$150d)	45	0.957	0.978	0.967
Blood	Post-pubertal ($\geq$150d)	33	0.969	0.939	0.954
Lung	Pre-pubertal ($<$150d)	102	0.960	0.931	0.945
Lung	Post-pubertal ($\geq$150d)	27	0.767	0.852	0.807

Table 3: Functional annotations of cross-species developmental markers. Annotations compiled from UniProt, GeneCards, and literature review. FC = $\log_2$ fold change (adult/infant); positive values indicate higher expression in adults, negative values indicate higher expression in infants. Pig FC from pig GTEx data (GEO: GSE257558); Human FC from Schaiter *et al.* (2024). Genes ordered by model importance within each functional module. "—" indicates the gene did not pass cross-species criteria (FDR < 0.10, $|\log_2 \text{FC}| > 0.5$) but is included based on its known biological role.

Gene	Developmental Function	Pig FC	Human FC
Module 1: Neuromuscular Maturation and Growth Signalling			
ACHE	Functions as the key enzyme at the neuromuscular junction and as a structural adhesion molecule guiding neuronal growth during development.	-3.34	-0.53
NRK	Embryonic Ste20-type kinase that promotes actin polymerisation via cofilin phosphorylation and activates JNK signalling during skeletal muscle formation; expression restricted to embryonic and early postnatal development.	-8.56	-2.45
KREMEN1	Negative regulator of Wnt/β-catenin signalling, partnering with Dkk1 to control cell survival and ensure proper patterning during postnatal development.	-1.69	-1.43
IGF2BP2	RNA-binding protein that promotes myoblast proliferation via translational control of growth-related transcripts (c-Myc, Igf1r, Sp1); postnatal downregulation required for terminal myogenic differentiation.	-2.19	-2.03
SORCS2	Molecular switch directing trafficking of signalling receptors to regulate the balance between neuronal growth and pruning during neuromuscular maturation.	-4.67	-0.95
ETV5	FGF-responsive PEA3-subfamily ETS transcription factor that promotes stem/progenitor cell self-renewal and myogenic differentiation in cooperation with MEF2; highly expressed during active postnatal myogenesis.	-2.80	-2.24
IGF2BP3	Oncofetal RNA-binding protein that promotes myoblast proliferation and differentiation via IGF2–PI3K/Akt signalling; pronounced postnatal silencing reflects the conserved transition from fetal growth programs.	-4.85	-1.23
CCDC8	Core component of the 3M complex (with CUL7 and OBSL1) that regulates microtubule dynamics, genome integrity, and growth hormone signalling during rapid postnatal growth; consistent with severe growth restriction in 3-M syndrome.	-4.44	-3.14
Module 2: Structural Refinement and ECM Remodelling			
TCAF1	Regulates TRPM8 ion channel trafficking and Ca^{2+} signalling; may support calcium-dependent processes during early postnatal muscle growth and metabolic maturation.	-4.72	-1.02
FGFRL1	Acts as a "decoy" receptor dampening excessive FGF signalling to ensure proper formation of critical structures including the diaphragm and skeleton.	-3.03	-2.10
WDR1	Cofactor of cofilin-mediated actin filament disassembly, essential for sarcomere assembly and myofibril organisation during postnatal muscle growth; higher expression in immature muscle reflects intensive cytoskeletal remodelling during myofibrillogenesis.	-1.69	-0.52
DLK1	Crucial regulator of the myogenic program that inhibits myoblast proliferation while promoting differentiation; high infant expression declines as the muscle fiber population matures.	-9.15	-2.88
FSCN1	Actin-bundling protein essential for formation of cellular protrusions during cell migration and tissue morphogenesis; highly expressed in embryonic somite-derived tissues, with postnatal downregulation as migratory programs complete.	-4.28	-1.95
POSTN	Secreted matricellular protein that mediates ECM remodelling, collagen fibrillogenesis, and connective tissue maturation during postnatal myofiber development; expression declines as tissue architecture stabilises.	-7.97	-2.93
FREM2	Basement membrane extracellular matrix protein of the FRAS/FREM complex maintaining epithelial–mesenchymal adhesion during organogenesis; postnatal downregulation reflects transition to collagen VII-dominated mature basement membranes.	-4.58	-1.05
ADAMTS2	Procollagen N-proteinase that cleaves propeptides from fibrillar procollagens to initiate collagen fibrillogenesis; high infant expression reflects intensive ECM biosynthesis during rapid postnatal growth.	-4.25	-1.83
COL11A2	Component of type XI collagen regulating collagen fibril organisation in cartilage and developing connective tissue; high infant expression reflects active skeletal morphogenesis and ECM assembly.	-8.35	-0.70
TRIM54	Muscle-specific E3 ubiquitin ligase (MuRF3) involved in sarcomere Z-disc assembly; expression increases postnatally as the mature contractile apparatus is established.	—	—
Module 3: Metabolic Transition			
ME1	Malic enzyme 1, catalysing oxidative decarboxylation of malate to pyruvate; a key source of NADPH for lipid biosynthesis. Expression increases during development, reflecting metabolic maturation.	2.59	0.64
MAOB	Mitochondrial outer membrane enzyme catalysing oxidative deamination of monoamines; postnatal upregulation reflects developmental maturation of oxidative metabolism and the glycolytic-to-oxidative fiber-type shift.	2.55	0.80
PLA2G4E	Calcium-dependent N-acyltransferase synthesising N-acylphosphatidylethanolamines (endocannabinoid precursors); species-discordant expression may reflect divergent intramuscular lipid metabolism programs.	2.32	-1.25

Table 4: Stage-by-stage expression trajectories of cross-species developmental markers. Mean TPM values across porcine developmental stages for genes with conserved pig-human developmental patterns. Stage boundaries: Infant (0–20 d), Early childhood (21–59 d), Pre-pubertal (60–149 d), Post-pubertal (150–365 d), Adult (>365 d). "Peak Change" indicates the developmental transition with the largest fold change. Genes are grouped by functional module and ordered by model importance within each module.

Gene	Infant	Early	Pre-pub	Post-pub	Adult	Peak Change
Module 1: Neuromuscular Maturation and Growth Signalling						
ACHE	17.5	7.6	3.4	1.3	1.7	Infant→Early
NRK	128.2	4.2	8.1	1.3	0.3	Infant→Early
KREMEN1	30.3	13.2	9.4	5.2	9.4	Infant→Early
IGF2BP2	11.3	1.5	1.9	1.0	2.5	Infant→Early
SORCS2	14.4	3.0	2.2	0.6	0.6	Infant→Early
ETV5	34.2	16.7	7.5	5.5	4.9	Early→Pre-pub
IGF2BP3	21.8	1.7	2.1	0.4	0.7	Infant→Early
CCDC8	177.4	28.2	25.4	13.4	8.2	Infant→Early
Module 2: Structural Refinement and ECM Remodelling						
TCAF1	78.9	8.3	8.1	3.8	3.0	Infant→Early
FGFRL1	77.4	15.6	14.2	4.3	9.5	Infant→Early
WDR1	202.9	125.6	95.0	51.9	63.1	Pre-pub→Post-pub
DLK1	338.8	3.9	6.4	0.7	0.6	Infant→Early
FSCN1	123.7	32.5	18.9	6.2	6.4	Infant→Early
POSTN	602.4	44.1	54.3	2.5	2.4	Pre-pub→Post-pub
FREM2	33.1	4.4	5.5	1.5	1.4	Infant→Early
ADAMTS2	51.1	16.6	16.0	6.4	2.7	Infant→Early
COL11A2	44.6	1.5	1.0	0.3	0.1	Infant→Early
TRIM54	78.3	351.1	149.4	377.9	144.4	Infant→Early
Module 3: Metabolic Transition						
ME1	2.8	4.0	3.9	6.9	16.9	Post-pub→Adult
MAOB	5.7	24.2	10.8	15.7	33.4	Infant→Early
PLA2G4E	0.1	1.4	0.7	0.6	0.5	Infant→Early

Table 5: Batch structure diagnostics across tissues. PERMANOVA R^2 quantifies the fraction of transcriptomic variance explained by each factor (999 permutations on PCA-reduced Euclidean distances). Cramér's V measures categorical association between developmental stage and batch variables (values approaching 1.0 indicate near-complete confounding). Silhouette scores measure clustering coherence in PCA space. TechBatch is a composite variable encoding sequencer generation and library layout (7 levels), designed to capture genuine technical variation without the structural confounding inherent in BioProject identity. BioProject shows high V ($>$0.87) because $\sim$90% of projects contain samples from a single developmental stage, making it structurally inseparable from Stage.

Tissue	Scheme	N	R^2(Stage)	R^2(TechBatch)	R^2(BioProject)	V(TechBatch)	V(BioProject)	Sil(Stage)
Muscle	4-class	761	0.171	0.170	0.730	0.395	0.871	0.036
Liver	4-class	309	0.060	0.147	0.697	0.315	0.954	0.038
Brain	2-class	43	0.175	0.395	0.764	0.840	0.870	0.284
Blood	2-class	78	0.052	0.166	0.506	0.365	0.973	0.074
Lung	2-class	129	0.029	0.201	0.824	0.563	1.000	0.078